

Assignment-1 [cite: 1]

Programming and Problem Solving with C [cite: 2]

Github: <https://github.com/dev-Sameer-2443/c-assignment/tree/C-Assignment/class-assignment-main>

Conditional & Looping Constructs [cite: 4]

1. Check Even or Odd [cite: 5]

```
ENTER NUMBER TO CHECK EVEN OR ODD: 26 [cite: 6]
```

```
NUMBER IS EVEN [cite: 7]
```

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2. Find Greatest of Three Numbers [cite: 9]

```
ENTER FIRST NUMBER : 24 [cite: 10]
```

```
ENTER SECOND NUMBER :13 [cite: 11]
```

```
ENTER THIRD NUMBER : 26 [cite: 12]
```

```
GREATEST NUMBER IS 26.000000 [cite: 13]
```

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3. Check for Leap Year [cite: 15]

```
ENTER YEAR: 2024 [cite: 16]
```

```
YEAR IS LEAP YEAR [cite: 17]
```

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4. Sum of Natural Numbers [cite: 19]

```
enter number to add till it: 5 [cite: 20]
```

```
sum of first 5 natural numbers is 15 [cite: 21]
```

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Conditional & Looping Constructs (cont.)

5. Multiplication Table [cite: 23]

```
ENTER NUMBER TO GET ITS TABLE: 9 [cite: 24]

9×1 =9 [cite: 25]
9 X 2=18 [cite: 26]
9 X 3=27 [cite: 27]
9 X 4=36 [cite: 28]
9 X 5=45 [cite: 29]
9 X 6=54 [cite: 30]
9 X 7=63 [cite: 31]
9 X 8=72 [cite: 32]
9 X 9=81 [cite: 33]
9 X 10=90 [cite: 34]
```

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6. Reverse a Number [cite: 36]

```
Enter any digit number: 1234

The reverse of 1234 is: 4321 [cite: 37]
```

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7. Simple Calculator (using switch) [cite: 39]

```
ENTER FIRST NUMBER:24 [cite: 40]

ENTER SECOND NUMBER:25 [cite: 41]
```

ENTER YOUR OPRETION TYPE- [cite: 42]

1 FOR ADDITION [cite: 43]

2 FOR SUBTRACTION [cite: 44]

3 FOR MULTIPLICATION [cite: 45]

4 FOR DIVISION [cite: 46]

1 [cite: 47]

SUM OF BOTH NUMBERS IS 49.000000 [cite: 48]

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Conditional & Looping Constructs (cont.)

8. Find Factorial of a Number [cite: 50]

```
ENTER NUMBER TO FINT FACTORIAL:5  
FACTORIAL IS 120 [cite: 51]
```

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9. Check for Palindrome Number [cite: 53]

```
Enter a number: 12321 [cite: 54]  
Number is palindrome. [cite: 55]
```

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Arrays (1D & 2D) [cite: 57]

1. Find Greatest in Array [cite: 58]

```
How many numbers do you want to compair:4 [cite: 59]  
1. ENTER NUMBER: 12 [cite: 60]  
2. ENTER NUMBER:54 [cite: 61]  
3. ENTER NUMBER: 2 [cite: 62]  
4. ENTER NUMBER:34 [cite: 63]  
54 is greatest among them [cite: 64]
```

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Arrays (1D & 2D) (cont.)

2. Calculate Sum and Average of Array Elements [cite: 66]

```
ENTER NUMBER OF ELEMENTS: 3 [cite: 67]

ENTER NUMBER 1: [cite: 68]
12 [cite: 69]

ENTER NUMBER 2: [cite: 70]
34 [cite: 71]

ENTER NUMBER 3: 54 [cite: 72]

SUM OF ALL NUMBERS 100 AND AVERAGE IS 33.000000 [cite: 73]
```

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3. Sort Array in Ascending Order [cite: 75]

```
Enter number of elements: 4 [cite: 76]

Enter 4 elements: [cite: 77]
12 [cite: 78]
54 [cite: 79]
11 [cite: 80]
9 [cite: 81]

Array in ascending order: [cite: 82]
9 11 12 54 [cite: 83]
```

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4. Count Frequency of Array Elements [cite: 85]

ENTER NUMBER OF ELEMENTS: 4 [cite: 86]

ENTER NUMBER 1: 1 [cite: 87]

ENTER NUMBER 2: 2 [cite: 88]

ENTER NUMBER 3: 1 [cite: 89]

ENTER NUMBER 4:4 [cite: 90]

1 COMES 2 TIMES [cite: 91]

2 COMES 1 TIMES [cite: 92]

4 COMES 1 TIMES [cite: 93]

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Arrays (1D & 2D) (cont.)

5. Merge Two Arrays [cite: 95]

```
ENTER NUMBER OF ELEMENTS IN FIRST ARRAY: 2
ENTER NUMBER OF ELEMENTS IN SECOND ARRAY: 3 [cite: 96]
FOR FIRST ARRAY
ENTER ELEMENT 1: 1 [cite: 97]
ENTER ELEMENT 2: 4 [cite: 98]
FOR SECOND ARRAY [cite: 99]
ENTER ELEMENT 1:3 [cite: 100]
ENTER ELEMENT 2: 2 [cite: 101]
ENTER ELEMENT 3:5 [cite: 102]
NEW ARRAY IS: [cite: 103]
1 [cite: 104]
4 [cite: 105]
3 [cite: 106]
2 [cite: 107]
5 [cite: 108]
```

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6. Add Two Matrices (2D Array) [cite: 110]

```
enter number of rows:2 [cite: 111]
enter number of columns:2 [cite: 112]
for first matrix [cite: 113]
enter number in (0,0):1 [cite: 114]
enter number in (0,1):2 [cite: 115]
```

```
enter number in (1,0):4 [cite: 116]
enter number in (1,1):3 [cite: 117]
for second matrix [cite: 118]
enter number in (0,0):7 [cite: 119]
enter number in (0,1):6 [cite: 120]
enter number in (1,0):5 [cite: 121]
enter number in (1,1):4 [cite: 122]
first matrix is- [cite: 123]
12 [cite: 124]
43 [cite: 125]
second matrix is- [cite: 126]
76 [cite: 127]
54 [cite: 128]
sum of both matrices is- [cite: 129]
88 [cite: 130]
97 [cite: 131]
```

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Arrays (1D & 2D) (cont.)

7. Find Transpose of a Matrix [cite: 133]

```
enter number of rows:2
enter number of columns:3 [cite: 134]
enter number in (0,0):1 [cite: 135]
enter number in (0,1):3 [cite: 136]
enter number in (0,2):2 [cite: 137]
enter number in (1,0):4 [cite: 138]
enter number in (1,1):6 [cite: 139]
enter number in (1,2):5 [cite: 140]
matrix is- [cite: 141]
132 [cite: 142]
465 [cite: 143]
transpose of matrix is- [cite: 144]
14 [cite: 145]
36 [cite: 146]
2.5 [cite: 147]
```

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8. Sum of Diagonal Elements of a Matrix [cite: 149]

```
enter number of rows/columns:2 [cite: 150]
enter number in (0,0):1 [cite: 151]
enter number in (0,1):9 [cite: 152]
enter number in (1,0):8 [cite: 153]
enter number in (1,1):4 [cite: 154]
```

matrix is- [cite: 155]

19 [cite: 156]

84 [cite: 157]

sum is 22 [cite: 158]

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Structures [cite: 160]

1. Student Details using Structure [cite: 161]

student's details- [cite: 162]

name: SAMEER KUMAR [cite: 163]

roll no.: 125113025 [cite: 164]

marks: 100 [cite: 165]

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2. Find Student with Maximum Marks [cite: 167]

NUMBER OF STUDENTS:3 [cite: 168]

FOR STUDENT 1 [cite: 169]

ENTER NAME OF STUDENT: SAMEER [cite: 170]

ENTER MARKS OF STUDENT:100 [cite: 171]

FOR STUDENT 2 [cite: 172]

ENTER NAME OF STUDENT: Vaibhav [cite: 173]

ENTER MARKS OF STUDENT: 90 [cite: 174]

FOR STUDENT 3 [cite: 175]

ENTER NAME OF STUDENT: Tanish [cite: 176]

ENTER MARKS OF STUDENT : 80 [cite: 177]

STUDENT WITH MAXIMUM MARKS [cite: 178]

NAME: SAMEER [cite: 179]

MARKS: 100 [cite: 180]

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Structures (cont.)

3. Book Details using Structure Array [cite: 182]

```
NUMBER OF BOOKS:3 [cite: 183]

FOR BOOK 1 [cite: 184]
ENTER NAME OF BOOK: MATHS [cite: 185]
ENTER PRICE OF BOOK:800 [cite: 186]
FOR BOOK 2 [cite: 187]
ENTER NAME OF BOOK: PHYSICS [cite: 188]
ENTER PRICE OF BOOK: 400 [cite: 189]
FOR BOOK 3 [cite: 190]
ENTER NAME OF BOOK: CHEMISTRY [cite: 191]
ENTER PRICE OF BOOK:10 [cite: 192]
BOOK NAME "MATHS" AND PRICE IS 800 [cite: 193]
```

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